

MEDICATION SAFETY AND HIGH-ALERT MEDICATIONS POLICY

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SYNTHETIC DEMONSTRATION DOCUMENT. Written for the CareFlow AI portfolio project; fictional policy.

1. Purpose

This policy reduces medication errors, with particular focus on high-alert medications, which carry a heightened risk of significant patient harm when used in error.

2. High-Alert Medication List

The following are designated high-alert medications at CareFlow General Hospital: all insulins (including insulin glargine and insulin lispro), anticoagulants (warfarin, apixaban, enoxaparin and unfractionated heparin), opioids (including morphine), concentrated potassium chloride, and intravenous dextrose solutions of 10% or more.

3. Prescribing Standards

Never use the abbreviation "U" for units; write "units" in full. Never use a trailing zero (write 5 mg, not 5.0 mg) and always use a leading zero (0.5 mg). Every prescription must include the dose, route, frequency and indication. Look-alike sound-alike (LASA) medicines are stored apart and displayed with tall-man lettering.

4. Independent Double Check

An independent double check by a second registered nurse or pharmacist is required before administering: any insulin infusion, any intravenous potassium, any opioid infusion or patient-controlled analgesia, and any intravenous anticoagulant.

5. Insulin Safety

Sliding-scale insulin used on its own is discouraged for inpatients; a basal-bolus regimen is preferred. Insulin doses are held or reduced when a patient is nil by mouth. Every hypoglycemia episode below 70 mg/dL must be reported and prompts a same-day dose review.

6. Anticoagulation Safety

6.1 Warfarin

Monitor the INR at least weekly during initiation and at least monthly once stable. The usual therapeutic range is INR 2.0 to 3.0. An INR above 4.5 is a critical value and must be reported to the prescriber immediately.

6.2 Apixaban

The standard dose for atrial fibrillation is 5 mg twice daily. Reduce to 2.5 mg twice daily when at least two of the following are present: age 80 years or older, body weight 60 kg or less, serum creatinine 1.5 mg/dL or higher.

6.3 Enoxaparin

The usual prophylactic dose is 40 mg subcutaneously once daily. When creatinine clearance is below 30 mL/min, reduce the prophylactic dose to 30 mg once daily.

7. Concentrated Potassium Chloride

Concentrated potassium chloride ampoules are removed from all ward stock and are available only from pharmacy. Use pre-mixed bags. The maximum infusion rate through a peripheral line is 10 mmol per hour.

8. Opioid Safety

Naloxone must be available wherever opioids are administered. Assess sedation and respiratory rate before each dose of an opioid.

9. Medication Reconciliation

Complete a medication reconciliation within 24 hours of admission and again at discharge. Verify allergies before the first dose of any new medication; a documented penicillin allergy must be checked before prescribing any beta-lactam antibiotic.

10. Reporting

All medication errors and near misses are reported through the incident reporting system within 24 hours. Reporting is non-punitive.